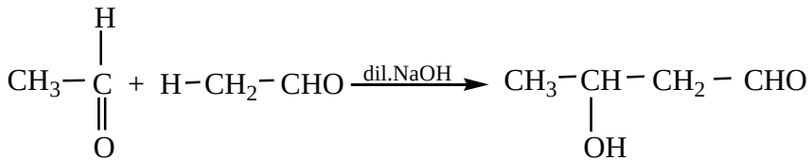
a) Acetaldehyde when warmed with dil NaOH gives β - hydroxyl butyraldehyde (acetaldol)



Acetaldehyde

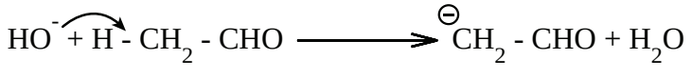
Acetaldol
(3 - Hydroxy butanal)

Mechanism

The mechanism of aldol condensation of acetaldehyde takes place in three steps.

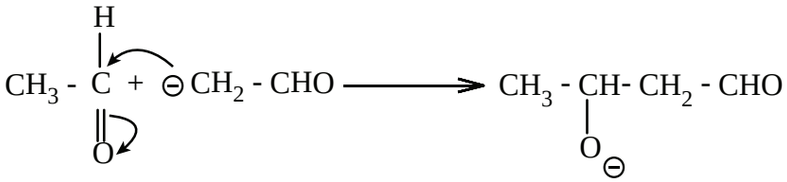
Step 1 :

The carbanion is formed as the α - **hydrogen atom** is removed as a proton by the base.



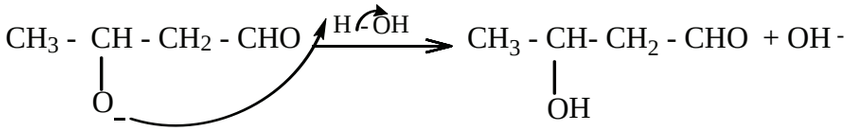
Step 2 :

The carbanion attacks the carbonyl carbon of another unionized aldehyde to form an alkoxide ion.



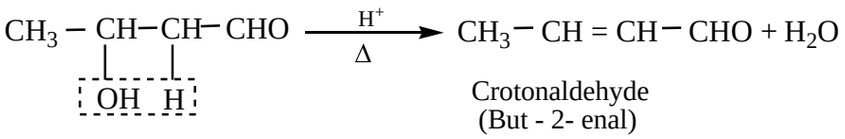
Step 3 :

The alkoxide ion formed is protonated by water to form aldol.



3-Hydroxy butanal

The aldol rapidly undergoes dehydration on heating with acid to form α - β unsaturated aldehyde.



Crotonaldehyde
(But - 2- enal)

ii) Crossed aldol condensation

Aldol condensation can also take place between two different aldehydes or ketones or between one aldehyde and one ketone such an aldol condensation is called **crossed or mixed aldol condensation.** This reaction is not very useful as the product is usually a mixture of all possible condensation products and cannot be separated easily.

XII U12-Carbonyl Compounds.indd 162 11-01-2021 16:10:38